

IMPHAL, India

WMO: 426230

Lat: 24.760N

Lon: 93.897E

Elev: 2540

StdP: 13.40

Time Zone: 5.50 (IND)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	41.6	43.6	38.2	37.1	49.9	39.9	39.8	49.8	10.3	66.8	9.0	66.5	0.2	140	0.256

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	90.0	75.2	88.7	75.2	87.5	75.0	79.0	86.6	78.1	85.5	77.4	84.4	5.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	155.0	83.9	76.1	149.9	82.6	75.4	146.4	81.7	45.0	87.0	44.0	85.9	43.2	84.8	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	9.8	7.8	37.3	93.6	1.5	1.4	36.2	94.6	35.3	95.4	34.5	96.2	33.4	97.2
			36.9	81.6	1.7	1.6	35.6	82.8	34.6	83.8	33.7	84.7	32.4	85.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.4	58.6	62.7	68.9	73.3	76.2	78.8	79.1	79.2	78.4	74.5	66.4	59.7
	DBStd	8.07	2.90	3.75	3.68	4.07	3.12	2.48	2.04	2.01	2.03	3.56	3.43	3.25
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	483	200	81	10	3	0	0	0	0	0	1	22	166
	CDD50	7794	267	356	585	700	812	863	901	904	853	759	493	301
	CDD65	2802	1	18	130	253	347	413	436	439	403	295	65	2
	CDH74	21460	68	375	1369	2081	2557	3206	3299	3329	2796	1820	506	54
	CDH80	5939	0	40	369	715	811	937	821	877	806	507	55	1
(14) Wind	WSAvg	2.1	1.4	2.1	2.6	2.7	2.3	2.3	2.4	2.6	2.1	1.5	1.3	1.3
(15) Precipitation	PrecAvg	81.4	0.6	1.9	3.2	5.7	10.2	14.1	14.8	12.4	9.7	5.7	2.1	0.6
	PrecMax	114.0	2.1	8.9	8.2	15.8	18.2	19.3	20.2	20.2	15.7	10.5	5.6	3.1
	PrecMin	47.1	0.1	0.5	0.4	1.5	4.6	8.0	9.3	8.0	5.6	1.7	0.8	0.0
	PrecStd	12.9	0.6	1.8	2.1	3.3	3.1	2.6	3.0	3.2	2.7	2.4	1.4	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	84.1	89.3	92.4	91.4	90.8	89.6	90.5	90.0	89.1	84.3	78.0
		MCWB	60.8	62.7	66.0	67.4	75.4	76.9	78.4	79.8	78.1	76.2	71.3	63.3
	2%	DB	75.9	80.9	86.4	89.2	88.6	88.9	87.7	88.3	88.5	87.2	81.6	75.3
		MCWB	59.6	61.9	65.5	69.6	74.1	76.5	77.5	78.6	77.2	75.0	68.6	61.9
	5%	DB	73.7	78.5	84.1	86.8	87.0	87.0	86.2	86.7	86.7	85.3	79.4	73.3
		MCWB	58.9	60.9	65.4	69.8	73.8	76.0	76.9	77.8	76.6	74.1	67.0	61.2
	10%	DB	71.3	76.1	81.6	84.4	84.8	85.3	84.6	84.9	84.7	83.0	77.1	70.9
		MCWB	57.9	60.1	64.8	69.9	73.2	75.6	76.5	77.2	76.0	73.0	65.7	59.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.7	66.5	70.7	74.6	77.4	79.0	79.3	81.1	79.3	78.2	72.2	65.5
		MCDB	73.2	77.3	82.0	83.9	87.2	87.7	87.1	88.2	87.9	86.0	81.8	73.9
	2%	WB	61.2	64.4	69.1	73.3	76.2	77.7	78.2	79.5	78.0	76.4	70.2	63.5
		MCDB	71.9	76.4	80.7	83.2	85.4	85.6	85.5	86.3	86.1	84.5	78.5	72.7
	5%	WB	59.7	62.8	67.7	72.3	75.1	77.0	77.4	78.3	77.1	75.1	68.9	62.0
		MCDB	71.1	74.8	79.8	81.9	83.9	84.4	84.2	85.0	84.8	83.0	76.2	71.2
	10%	WB	58.5	61.2	66.3	71.1	74.1	76.2	76.8	77.4	76.4	74.0	67.4	60.7
		MCDB	69.8	73.5	77.5	80.4	82.1	83.1	83.7	83.3	83.3	81.0	74.1	69.3
(35) Mean Daily Temperature Range	MDBR		29.6	29.4	26.3	20.4	16.6	12.7	11.1	11.8	13.5	17.6	24.5	26.6
	5% DB	MCDBR	31.7	32.3	30.1	25.5	20.5	15.8	13.8	14.8	16.7	20.7	26.2	28.1
		MCWBR	16.6	15.3	12.5	10.0	8.6	6.3	5.5	7.0	7.2	9.4	13.6	15.5
	5% WB	MCDBR	29.7	28.8	24.9	19.9	17.5	13.4	12.4	13.7	14.9	17.5	22.0	25.9
		MCWBR	16.3	14.8	12.4	9.6	8.3	6.0	5.5	7.0	6.7	8.3	12.3	14.9
(40) Clear-Sky Solar Irradiance	taub		0.393	0.438	0.592	0.572	0.547	0.498	0.435	0.446	0.454	0.423	0.370	0.356
	taud		2.230	2.109	1.735	1.849	1.975	2.179	2.377	2.332	2.283	2.321	2.387	2.389
	Ebn at Noon		271	266	231	240	245	255	272	269	265	268	278	280
	Edh at Noon		40	48	73	67	59	48	39	41	42	38	34	33
(44) All-Sky Solar Radiation	RadAvg		1314	1528	1649	1671	1567	1382	1322	1373	1360	1351	1344	1213
	RadStd		84	92	131	170	134	122	95	86	98	84	102	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page